LoanGenie

Loan Approval Predictor

Logistic Regression • Streamlit • Docker

Overview

LoanGenie predicts loan approval eligibility based on applicant financial profile. Built on 4,269 real loan records, it uses a **Logistic Regression** model with engineered features achieving **98.24% test accuracy** and **ROC-AUC of 0.993**.

Tech Stack

Category	Tool	Purpose
Modelling	scikit-learn	Logistic Regression, GridSearchCV
Data	pandas, numpy	EDA, Feature Engineering
Scaling	StandardScaler	Feature normalisation
App	Streamlit	Interactive UI
Deployment	Docker	Containerised inference

Dataset

- 4,269 loan records with 13 features including CIBIL score, income, assets, loan amount and term
- Target: **Approved (62%)** / **Rejected (38%)** — mild imbalance, handled via Stratified Split
- No missing values; asset columns aggregated from 4 sub-categories into single **assets** feature

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Feature Engineering

Six techniques applied to expose non-linear CIBIL behaviour to the linear model:

- **CIBIL Ordinal Bins** — 9 ordered buckets mirroring real bank lending policy thresholds
- **Threshold Flags** — binary indicators for CIBIL ≥ 700, ≥ 750, < 500
- **Quadratic Term** — cibil_score² to capture curved approval response
- **Financial Ratios** — loan-to-income, loan-to-assets, income-per-dependent
- **Interaction Feature** — CIBIL × income to jointly encode creditworthiness + capacity
- **Log Transforms** — log1p applied to income, loan amount and assets to reduce right-skew

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Model & Evaluation

- Algorithm: **Logistic Regression** (solver: lbfgs, L2 regularization)
- Tuning: **GridSearchCV** over C ∈ {0.001 → 200} with **5-fold StratifiedKFold**
- Best C = 50 | CV Accuracy: **97.28%** | Test Accuracy: **98.24%**
- ROC-AUC: **0.993** | F1-Score: **0.98** on both classes | Precision = Recall (FP ≈ FN)
- No overfitting — train/test gap < 0.5%; confirmed via **Learning Curve** and **Confusion Matrix**

Project Structure

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LoanGenie/ ├── loan_approval_app.py     ← Streamlit app ├── Loan_Approval.ipynb      ← Full notebook ├── model.pkl                ← Trained LR model ├── scaler.pkl ← Fitted StandardScaler ├── Dockerfile ├── requirements.txt └── loan_approval_dataset.csv
```

Run Locally

Without Docker

pip install -r requirements.txt

streamlit run loan_approval_app.py

With Docker

docker build -t loangenie .

docker run -p 8501:8501 loangenie

Open **http://localhost:8501** in your browser.
